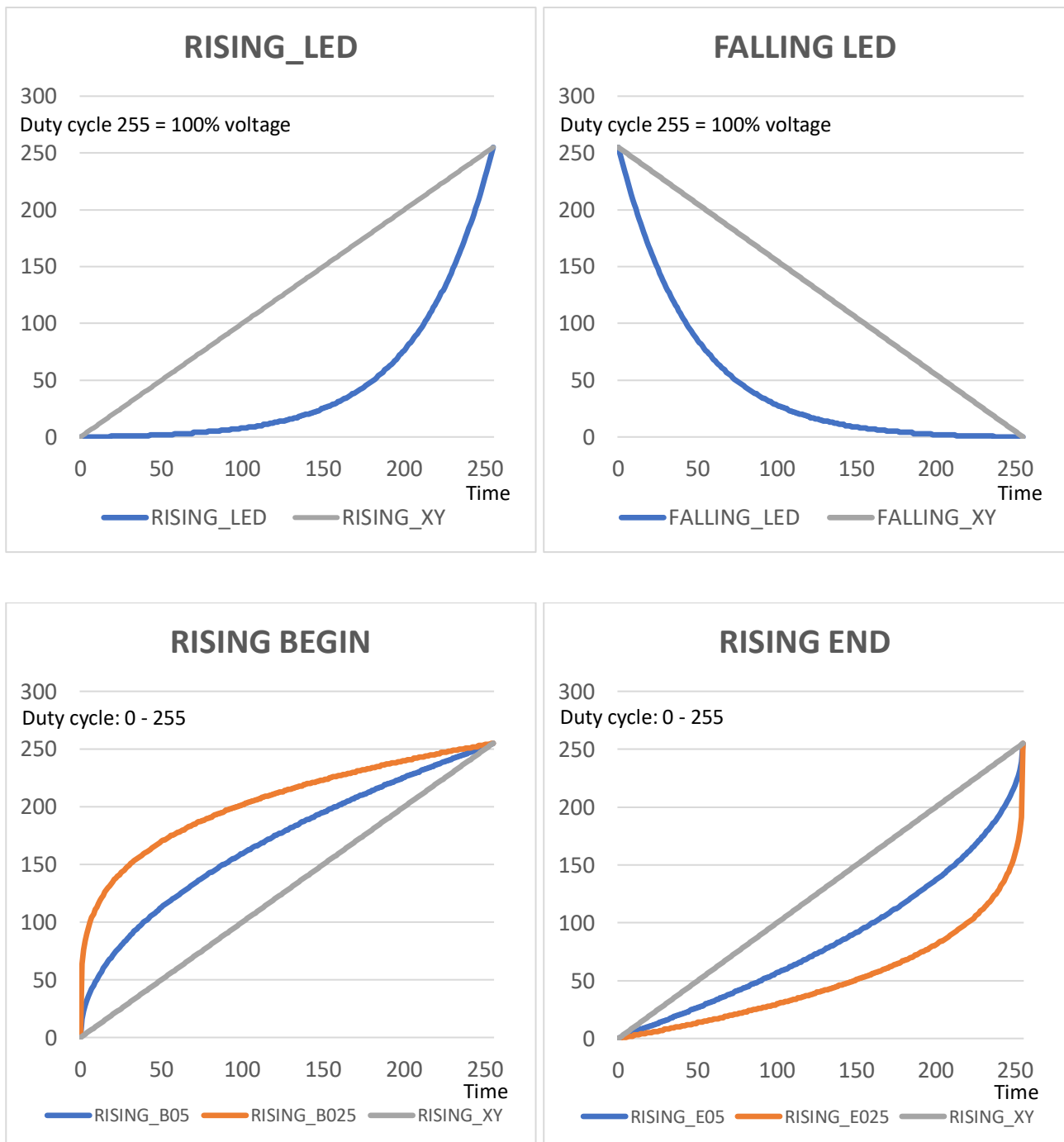


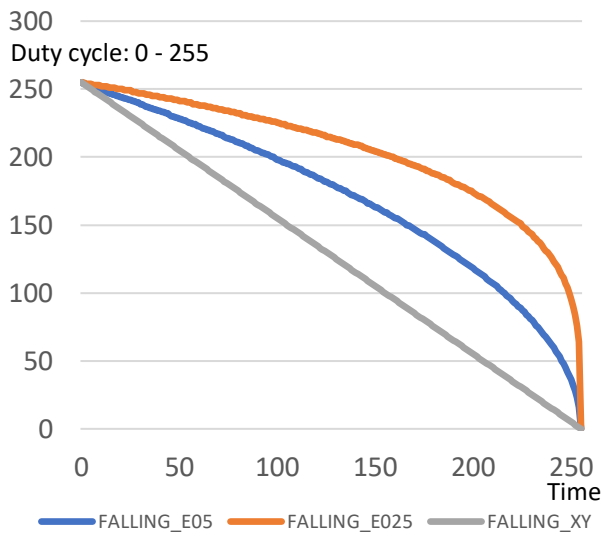
PWM_curves

MTD2A: Model Train Detection And Action – **Arduino** library <https://github.com/MTD2A/MTD2A>
Jørgen Bo Madsen / V1.2 / 17-07-2026

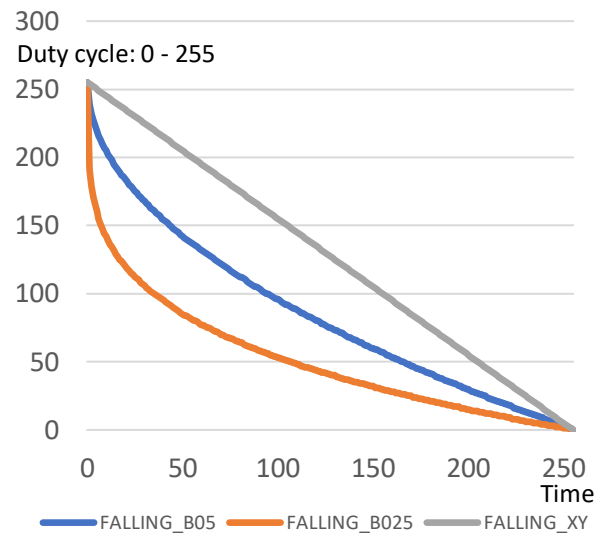
24 different mathematical [PWM](#) curves are designed: 2 linear, 4 exponential, 10 power, 4 Bézier and 4 Sine functions. The functions are intended to smooth out physical conditions that have a nonlinear relationship between time and function. E.g. LED fade in/fade out and acceleration and deceleration DC motors (trains) and servo.



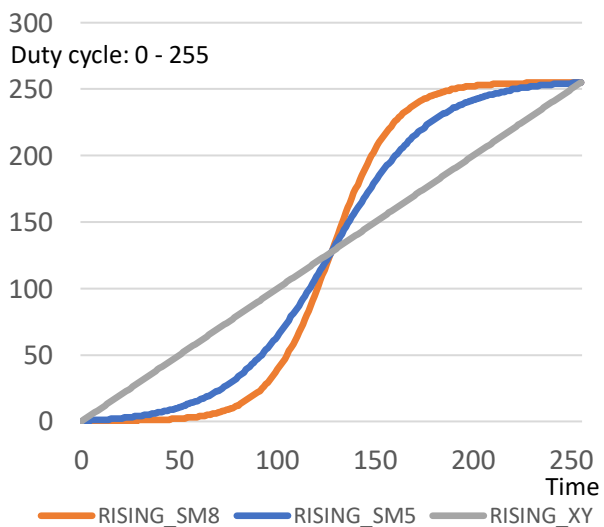
FALLING END



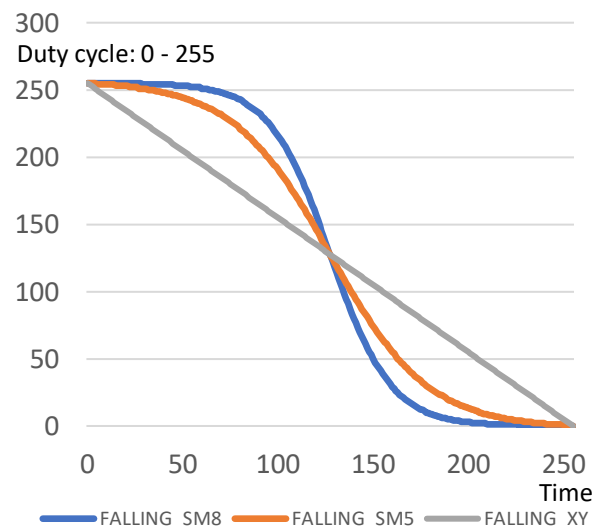
FALLING BEGIN



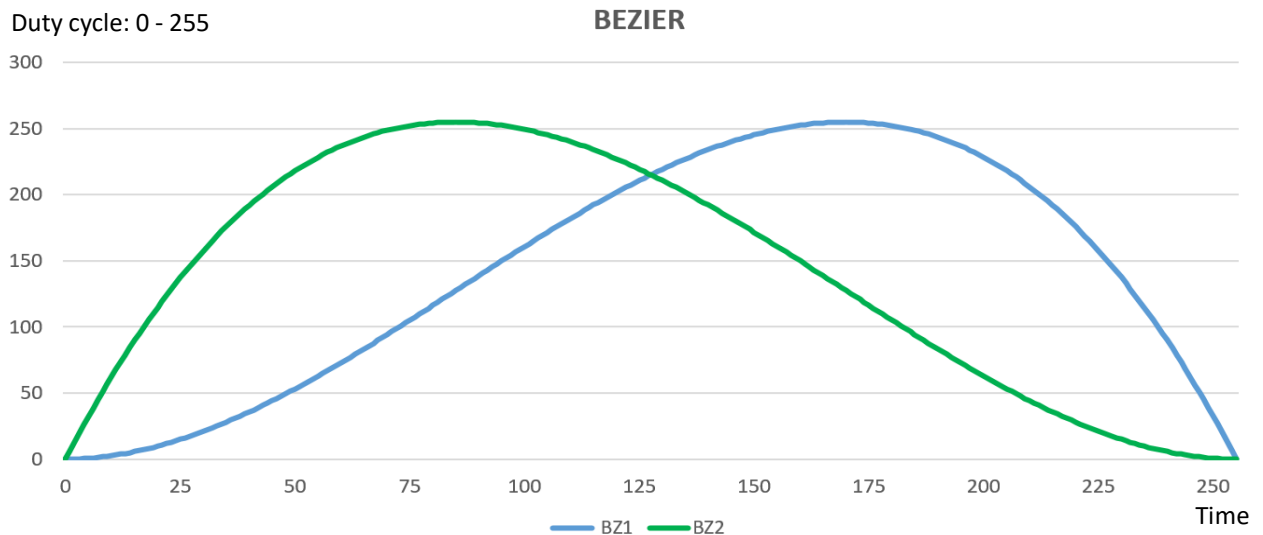
RISING SM-CURVE

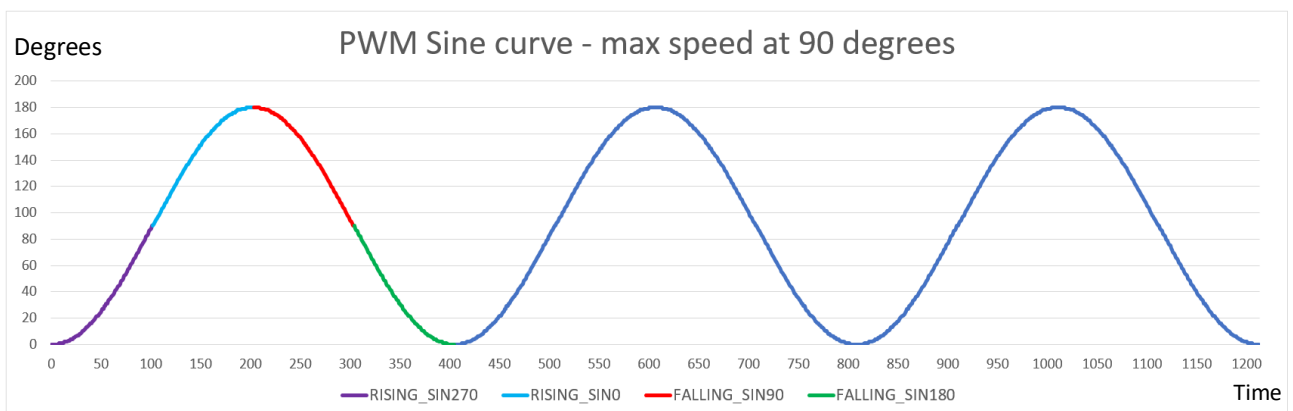
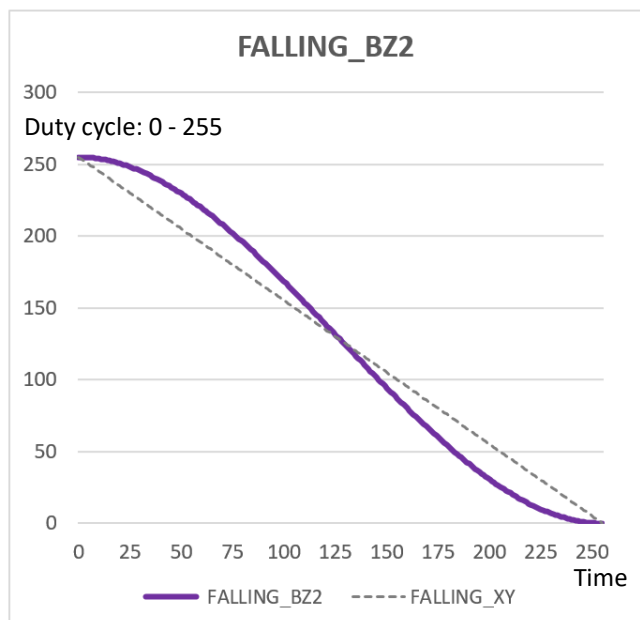
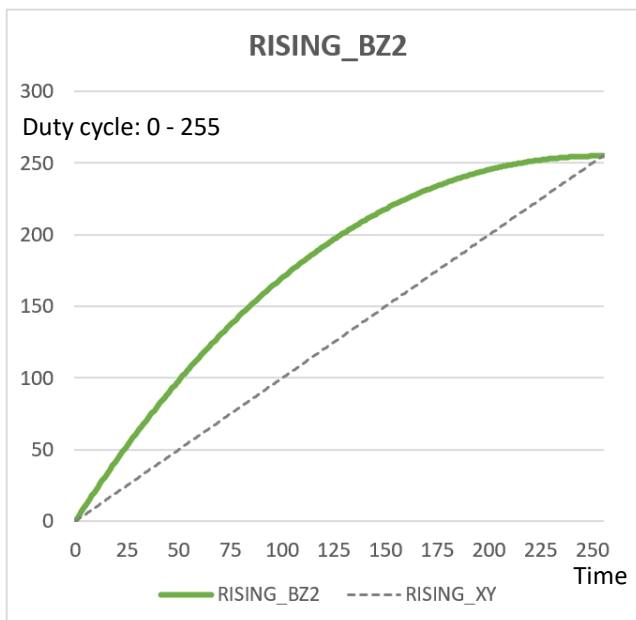
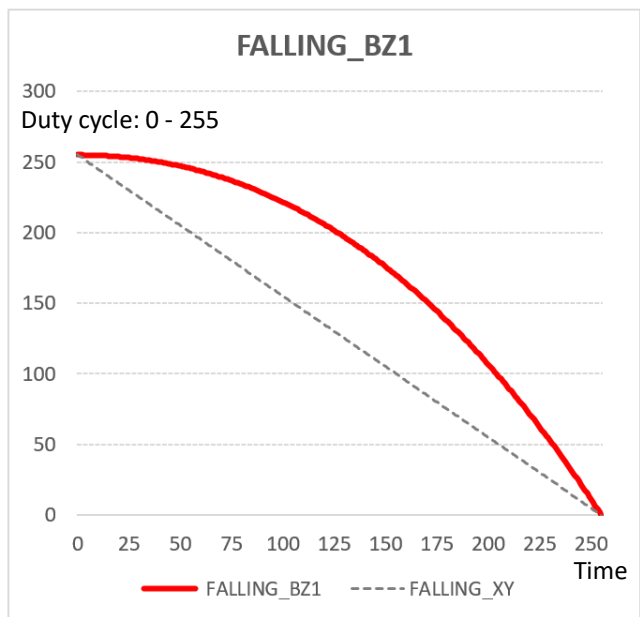
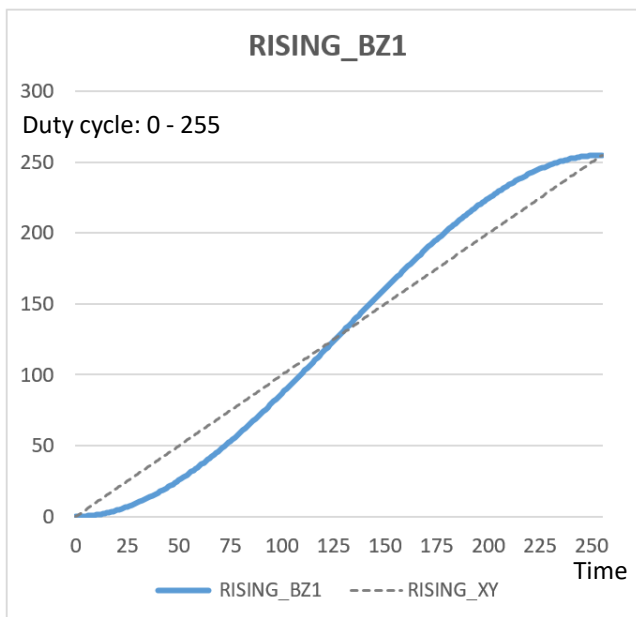


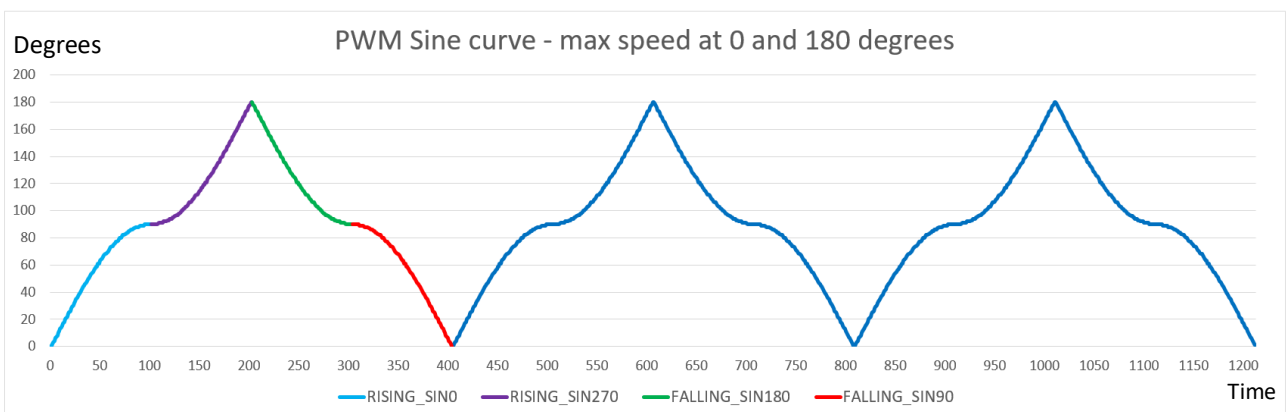
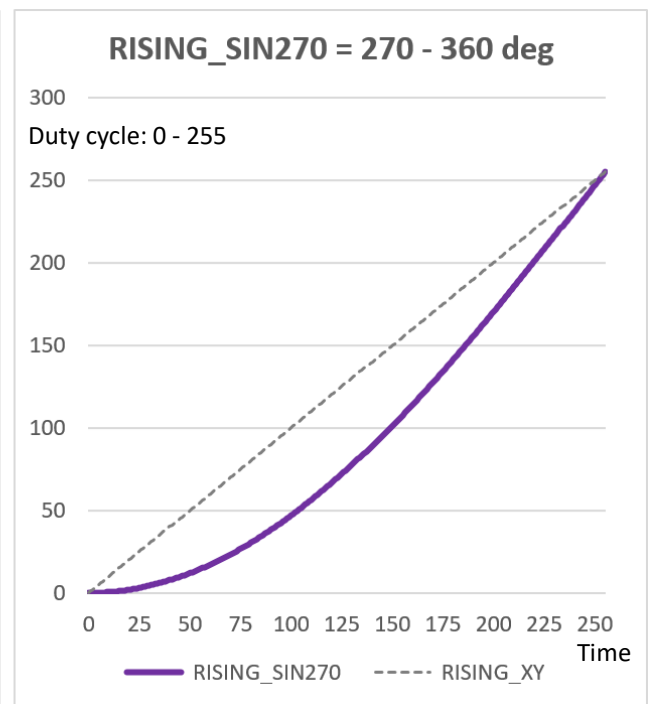
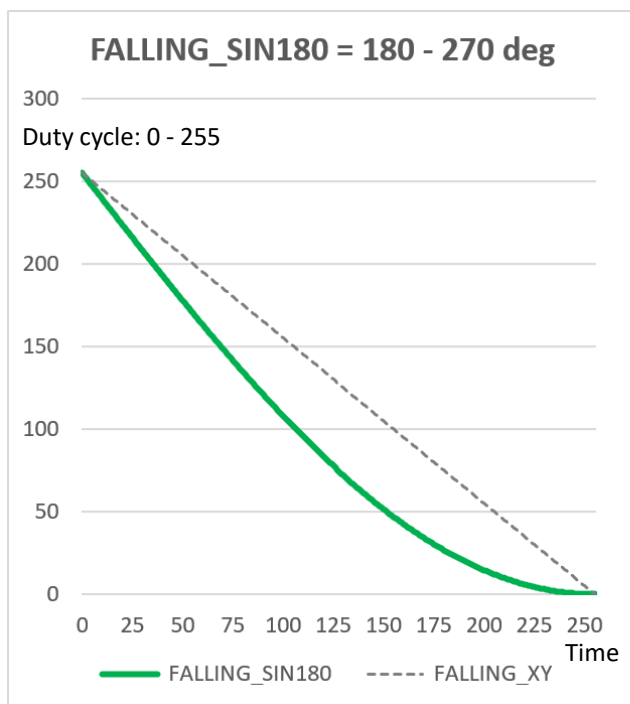
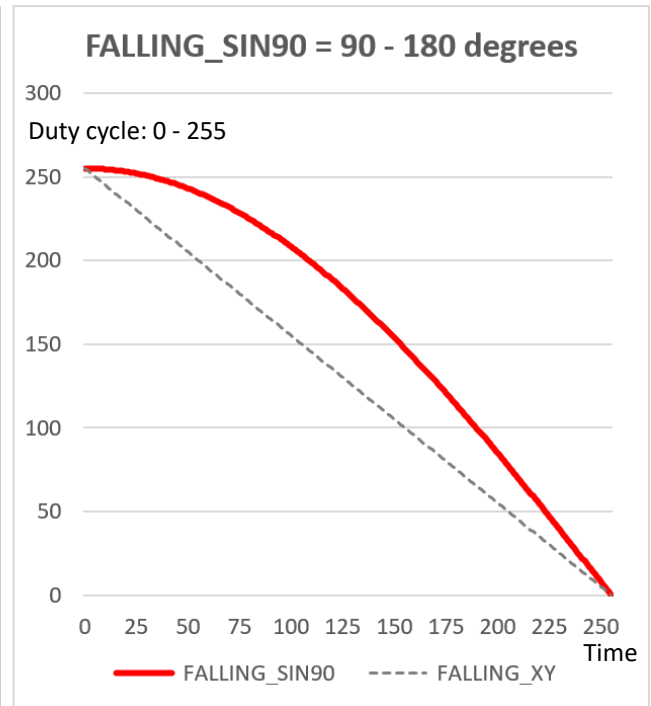
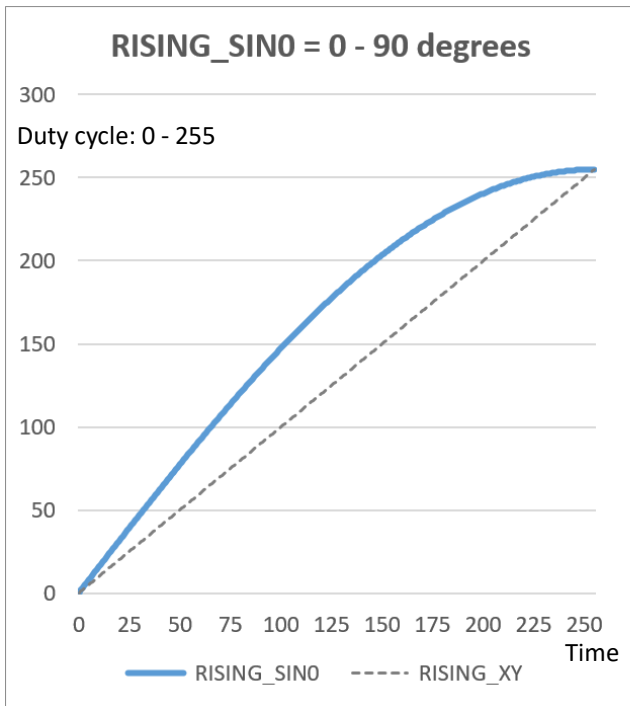
FALLING SM-CURVE

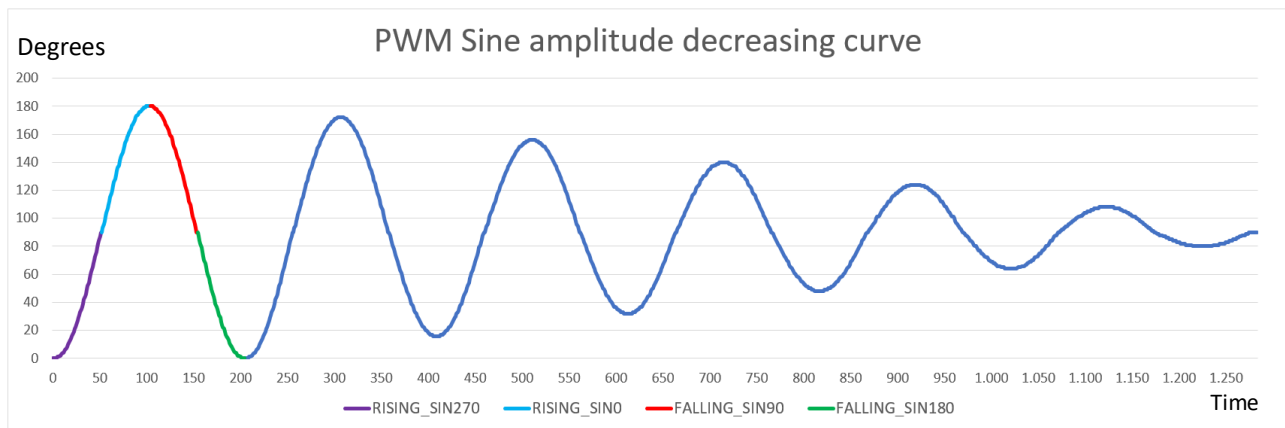


BEZIER









The various curves are named as global constants (MTD2A_const.h):

MIN_PWM_VALUE

MAX_PWM_VALUE

NO_CURVE

RISING_XY

RISING_LED

RISING_SM8

RISING_B05

FALLING_B05

RISING_BZ1

RISING_SIN0

FALLING_XY

FALLING_LED

RISING_SM5

RISING_B025

FALLING_B025

FALLING_BZ1

FALLING_SIN90

FALLING_SM8

RISING_E05

FALLING_E05

RISING_BZ2

RISING_SIN270

FALLING_SM5

RISING_E025

FALLING_E025

FALLING_BZ2

FALLING_SIN180

Clarification: [Pulse-width modulation - Wikipedia, the free encyclopedia](https://en.wikipedia.org/wiki/Pulse-width_modulation)